

Kaung Nyo Lwin

AI Engineer & Researcher

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PROFILE

AI engineer building computer vision and generative AI applications, from model training and evaluation to cloud deployment. Experience spans retrieval-augmented generation, LLM fine-tuning, biometric verification, and intelligent video analytics, backed by Python engineering and solution architecture.

TECHNICAL SKILLS

Generative AI: LLM fine-tuning, LoRA, GRPO, RAG, embeddings, semantic search; Hugging Face Transformers, TRL, LangChain, LlamaIndex, OpenAI API, FAISS.

Computer vision: PyTorch, OpenCV, YOLO, ByteTrack, ONNX, TensorFlow Lite; object detection, multi-object tracking, biometrics, model evaluation.

ML engineering and cloud: Python, SQL, FastAPI, Docker, Google Cloud, BigQuery, Vertex AI, Terraform, PostgreSQL, Git; data pipelines and cloud deployment.

AI & MACHINE LEARNING EXPERIENCE

CV Researcher

Jul 2026–Present

AIT × Modular Open Source Identity Platform (MOSIP)

Biometric AI · Face liveness and anti-spoofing

- Develop deep learning workflows for face anti-spoofing and liveness detection, supporting biometric verification for MOSIP digital identity.
- Engineer PyTorch training pipelines and model evaluation across ONNX and TensorFlow Lite, with demographic analysis and export for mobile SDK integration.

CV Engineer

Jul 2026–Present

AIT × CDG Group

Driver Licensing Management System

- Engineer a computer vision system for driving-test assessment, turning camera footage into calibrated vehicle trajectories and reviewable test evidence.
- Build OpenCV pipelines for camera calibration, perspective correction, and marker tracking, with RTSP capture and browser tools for field setup and validation.

Research Engineer

Mar 2026–Present

AIT × Toyota Mobility Foundation

TRUST · Intelligent transport and road safety

- Build a cloud analytics platform that combines vehicle telemetry and CCTV analysis to help researchers investigate traffic behavior and road-safety risks.
- Develop object detection and multi-object tracking pipelines with YOLO and ByteTrack, connecting vehicle trajectories to geospatial analysis.
- Deploy FastAPI services and BigQuery analytics on Google Cloud using Docker and Terraform, delivering authenticated dashboards for research teams.

LLM Developer Intern

Jul 2025–Oct 2025

AI Brain Lab

Generative AI · Document intelligence

- Developed generative AI applications for ESG reporting and financial-document question answering using LLMs, retrieval-augmented generation (RAG), and FastAPI.
- Built document ingestion, table extraction, and embedding pipelines for PDF, Excel, and CSV sources, enabling semantic search across Stock Exchange of Thailand materials.

RESEARCH

L2BC: Learning with Verifiable Rewards

Asian Institute of Technology | Completed master’s thesis.

- Developed a reinforcement learning framework using GRPO and LoRA to train compact LLMs for text-to-visualization with rule-based, verifiable rewards.
- Built verifiers that execute generated code, reconstruct chart type and data, and check correctness against ground truth; used this feedback to adapt training.
- Demonstrated single-pass generation competitive with larger agentic pipelines on VisEval, with ablation and generalization studies to identify what the verifier teaches.

SOLUTION ARCHITECTURE & EARLIER EXPERIENCE

Solutions Architect (Contract)

Aug 2026–Present

MA’HAR EMPEROR (Audit Firm)

Project Management & Tracking System

- Own end-to-end delivery of a project management system, from requirements and data collection through solution architecture, development, and production rollout.
- Build Odoo/PostgreSQL workflows for project planning, timesheets, access control, and management reporting, translating audit operations into a maintainable business system.

Senior Business Analyst

Feb 2023–Jul 2024

KEYfields

- Translated accounting and logistics requirements into SQL data models, stored procedures, and reporting workflows for SaaS products.
- Automated deployments and database backups with Python and PowerShell, improving operational reliability and reducing manual effort.

Senior Asset Management Analyst

May–Sep 2020

Telecom International Myanmar

Previously Asset Management Analyst · Jun 2017–Apr 2020

- Built Excel/VBA reporting tools and KPI dashboards to streamline asset tracking, inventory planning, and telecom rollout coordination.

Telecom Installation Engineer

Feb–May 2016

Ericsson Myanmar

- Delivered site surveys, equipment configuration, and network testing for telecom installations.

EDUCATION

Master of Engineering, Data Science and Artificial Intelligence

2024–2026

Asian Institute of Technology, Thailand

Bachelor of Engineering, Avionics

2010–2015

Myanmar Aerospace Engineering University, Myanmar

CERTIFICATIONS & SKILLS

IBM: Data Science, Data Analyst. DeepLearning.AI: Machine Learning, Deep Learning, TensorFlow Developer, Mathematics for Machine Learning and Data Science. University of Illinois: Fundamentals of Accounting.

Additional tools: scikit-learn, NumPy, pandas, MLflow, Power BI, SQL Server, PowerShell, Excel/VBA, Odoo.

Languages: Myanmar (native); English (C1).